

6.3 Video Correction

In Video 6.3 Convolution Theorem, from 2:40 to 2:55, in the verbal explanation of the derivation, instead of replacing u by $u + \tau - \tau$, it should be replacing x by $u + \tau - \tau$.

Convolution and Fourier Transform

$$\text{Convolution: } g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(\tau)h(x - \tau) d\tau$$

Fourier Transform of $g(x)$:

$$G(u) = \int_{-\infty}^{\infty} g(x)e^{-i2\pi ux} dx$$

$$G(u) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau)h(x - \tau)e^{-i2\pi ux} d\tau dx$$

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